Code No: 115EM **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August/September - 2024
SOFTWARE ENGINEERING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Distinguish PSP and TSP.	[2]
b)	List and define CMMI levels.	[3]
c)	Which characteristics are needed for a good SRS document?	[2]
d)	Define user requirements and system requirements.	[3]
e)	What are the adornments in the UML?	[2]
f)	What are tagged values and constraints?	[3]
g)	Give an example for regression testing.	[2]
h)	List the metrics for source code.	[3]
i)	Define software reliability.	[2]
j)	How does software reviews affect the software development?	[3]

PART – B

(50 Marks)

2.a) Define the term software. Describe its various characteristics of software.
b) Give the layered technology used in software development process. [5+5]

OR

3.a) Explain about Incremental Process Model.
b) In detail, discuss about Process Patterns and Process Assessment. [5+5]

4.a) What is the role of feasibility study in requirement engineering process? How it is performed?
b) What is the importance of user requirements? Explain. [5+5]

OR

5.a) In brief discuss about structural model with an example? What are its advantages?
b) Give an overview of behavioral models using an example. [5+5]

6.a) How design process is useful in development process? What are different design quality guidelines that lead to a good design?
b) Describe pattern based software design. [5+5]

OR

7.a) What is Software architecture? Analyze various architectural styles and patterns.
b) How interface affect the software usage? Mention various interface design steps. [5+5]

8.a) How does White Box testing is different from Black box testing?
b) Discuss the test strategies for conventional software. [5+5]

OR

9.a) Give the Frame work for product metrics.
b) What are the metrics used for software quality? Briefly describe. [5+5]

10.a) Distinguish between Reactive Vs Proactive risk strategies.
b) Elaborate on risk refinement. [5+5]

OR

11.a) What are the different statistical software quality assurance approaches? Explain.
b) How Formal Technical reviews are conducted? Discuss in detail. [5+5]

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